

BEAT-Unet: *Boundary-Enhanced Attention Transformer U-Net*

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Abstract

Salient object segmentation aims to identify and accurately delineate visually prominent regions in an image, a task essential for numerous downstream applications such as object recognition, image editing, and robot perception. In this paper, we propose a deep encoder-decoder framework that integrates region-level representation learning with boundary-preserving cues to enhance both global consistency and local detail accuracy. To further improve optimization stability and segmentation quality, we employ a hybrid loss function combining Binary Cross-Entropy and Dice loss, enabling the model to effectively balance pixel-wise prediction fidelity with structural completeness. The proposed method is trained on the DUTS-TR dataset and evaluated on the ECSSD benchmark. Experimental results demonstrate strong performance across standard saliency metrics, achieving high precision, accurate object localization, and robust generalization to complex scenes with low contrast and heavy background clutter. Qualitative comparisons further confirm the model's capability in producing sharp and coherent salient regions. These findings highlight the effectiveness of the proposed approach and its potential for broader applications in visual attention modeling.

I. Introduction

Visual Saliency Segmentation is a fundamental research direction in computer vision, aiming to model the human visual system's ability to focus on the most informative regions of a scene. The concept of visual

saliency was originally grounded in biologically inspired attention models, notably the work of Itti, Koch, and Niebur (1998) [2], which demonstrated that a combination of low-level visual features, such as color, intensity, and orientation, can predict regions that attract human attention. Subsequent studies, including comprehensive reviews by Borji and Itti, as well as benchmark datasets such as MSRA-B [1], ECSSD [3], and DUT-OMRON [2], have established standardized evaluation protocols for this problem, facilitating rigorous comparison among methods.

With the advent of deep learning, saliency segmentation has achieved higher accuracy through models such as Fully Convolutional Networks (FCNs), U-Net, and spatial channel attention mechanisms. Recent approaches leveraging deep encoder-decoder architectures, multi-stream networks, or self-attention mechanisms have demonstrated powerful contextual representation learning, enabling precise segmentation of salient regions even in complex and cluttered scenes. These advances not only improve the quality of saliency maps but also expand their applicability to tasks such as object recognition, video summarization, robotic navigation, and medical image analysis.

Moreover, contemporary research emphasizes **boundary preservation and structural consistency**, often integrating edge-aware modules or hybrid loss functions to enhance the delineation of salient objects. Such strategies, combined with multi-scale and transformer-based context modeling, have shown promising improvements in both quantitative metrics and qualitative visual quality.

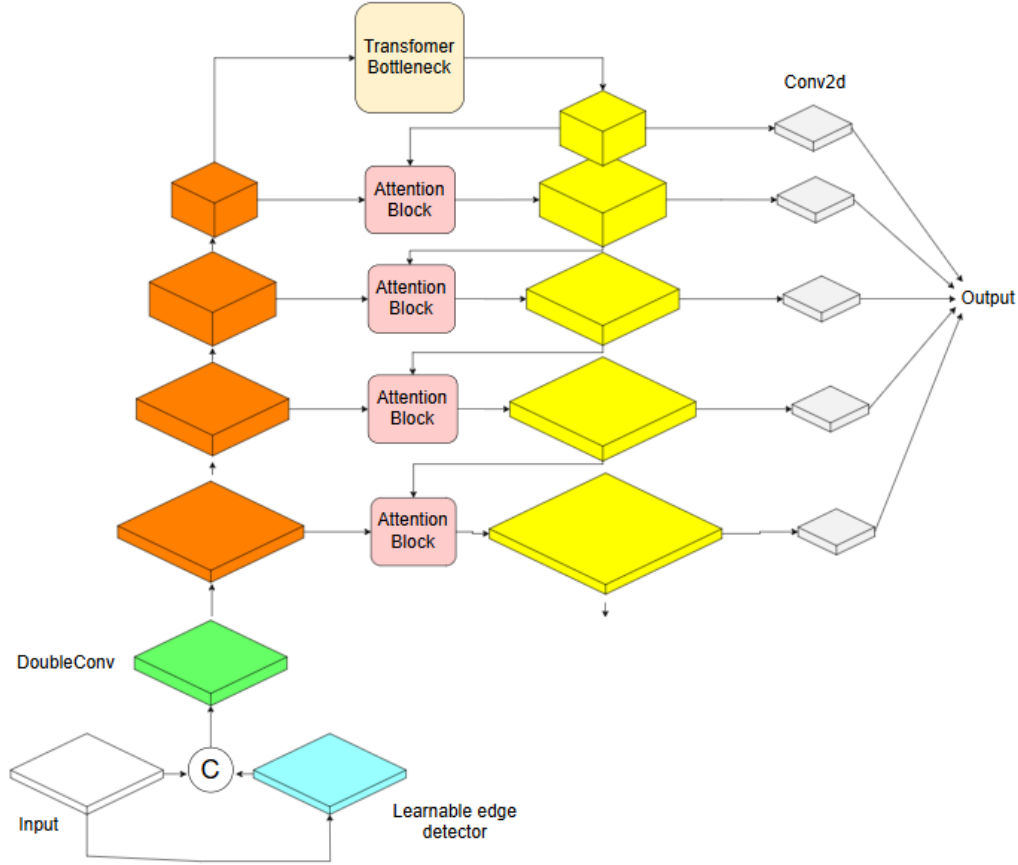


Figure 1: The proposed method: BEAT-NET

II. Method

The proposed framework is built upon an enhanced U-Net architecture integrating three major components: (i) a learnable edge detection module, (ii) an efficient transformer bottleneck with patch embedding, and (iii) attention-gated skip connections. This design aims to improve structural awareness, spatial consistency, and long-range contextual modeling for saliency-oriented segmentation.

A. Learnable Edge Detection

To enhance the structural cues provided to the encoder, we introduce a Learnable Edge Detector placed directly at the input stage. The module consists

of a single 3×3 convolutional filter followed by Instance Normalization and a Sigmoid activation. Unlike hand-crafted Sobel or Laplacian operators, this learnable kernel adapts to the training data, producing a normalized edge map that captures high-frequency boundary information. The resulting edge map is concatenated channel-wise with the original input, forming a richer representation for the first convolutional block.

B. Encoder Architecture

The encoder follows the standard U-Net design with repeated DoubleConv blocks, each comprising two sequences of convolution, batch normalization, ReLU layers. Spatial resolution is progressively reduced us-

ing MaxPool operations, enabling hierarchical feature extraction with increasing receptive fields. The output of each encoder stage is forwarded to the decoder through skip connections after being processed by attention gates (Section II-D).

C. Transformer Bottleneck with Patch Embedding

To capture long-range dependencies, the bottleneck replaces conventional convolution blocks with a Transformer Encoder. The input feature map is first downsampled using a patch-embedding convolution with a stride equal to the patch size, effectively reducing the number of tokens by a factor of P^2 . Flattened patches are enriched with sinusoidal positional encodings, allowing the transformer to preserve spatial ordering.

A lightweight transformer stack with multi-head self-attention is employed to model global contextual relations. The output tokens are reshaped back to a spatial representation and upsampled using a transposed convolution to restore the bottleneck resolution. This design significantly reduces computational cost while retaining the contextual learning strengths of transformer architectures.

D. Attention-Gated Skip Connections

To prevent irrelevant encoder features from propagating to the decoder, we incorporate Attention Gates (AGs). Each AG receives:

- A gating signal from the decoder representing high-level semantic cues.
- A skip-connection feature map from the encoder representing localized low-level details.

The two signals are projected to a common intermediate dimensionality and fused additively. A learned attention coefficient is then generated via a 1×1 convolution followed by a Sigmoid activation. The resulting pixel-wise attention map suppresses

background noise while enhancing salient structures. This mechanism ensures that only task-relevant features contribute to the reconstruction process.

III. Experiment

A. Datasets

To evaluate the effectiveness of the proposed salient object segmentation framework, we conduct experiments on two widely adopted benchmark datasets.

DUTS [4] is the largest publicly available dataset for saliency detection, consisting of 10,553 training images (DUTS-TR) and 5,019 test images (DUTS-TE). Following common practice, we perform training exclusively on the DUTS-TR split, which contains diverse scenes with complex background structures and multiple-object compositions.

ECSSD [3] contains 1,000 high-quality natural images with rich semantic structures and significant foreground-background contrast. Despite its moderate size, ECSSD is considered a highly challenging dataset due to the presence of irregular salient regions and heavy background clutter. This dataset is used solely for testing in our evaluation protocol.

B. Implementation Details

All experiments are implemented using PyTorch and trained on a single NVIDIA GPU. Images are resized to 256×256 during training and testing unless otherwise specified. Standard data augmentation techniques, including random horizontal flipping, color jittering, and random cropping, are applied to improve generalization.

We train the model using the Adam optimizer with a learning rate of 1×10^{-4} and weight decay of 5×10^{-4} . The batch size is set to 8, and training is performed for 40 epochs. The loss function combines binary cross

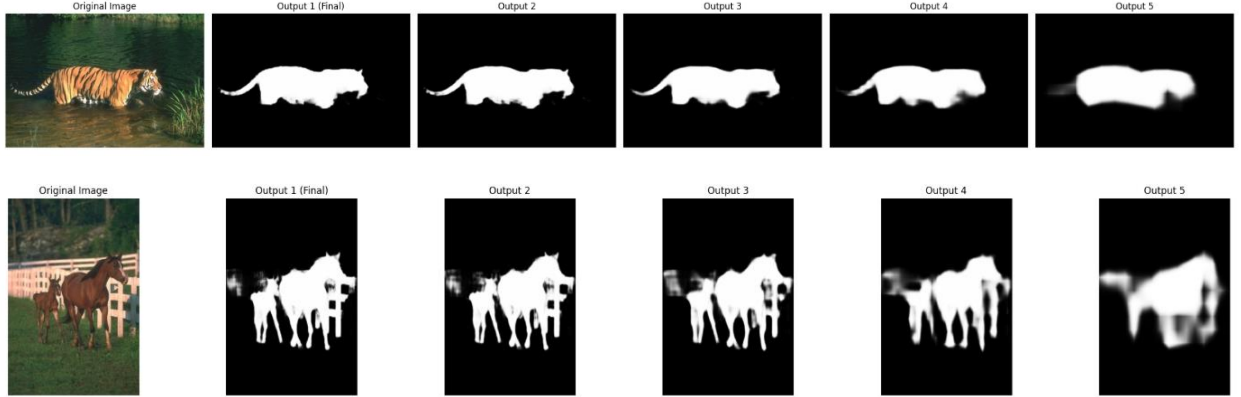
Table 1: Performance Metrics of the BEAT-NET on ECSSD Dataset

Metrics	MAE ↓	$F_\beta \uparrow$	F1-Score↑	S-measure↑
LCNN[5]	0.16	0.71	-	-
KRS[6]	0.133	0.801	-	-
PAGE-Net [7]	0.042	0.924	-	-
SAM2-Unet [8]	0.020	-	-	0.950
Ours	0.0585	0.8898	0.8711	0.9410

Table 2: Performance Metrics of the BEAT-NET on DUTS-TE Dataset

Metrics	MAE ↓	$F_\beta \uparrow$	F1-Score↑	S-measure↑
KRS[6]	0.123	0.66	-	-
PAGE-Net [7]	0.051	0.815	-	-
SAM2-Unet [8]	0.020	-	-	0.934
Ours	0.0687	0.7600	0.7740	0.9309

Figure 2: Visualize the outputs from all layers of BEAT-NET.



Entropy, MAE, and Dice loss to balance pixel-wise accuracy and region-level overlap.

C. Evaluation Metrics

We employ three standard metrics for quantitative comparison:

1. Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |P_i - G_i|$$

2. F-measure (F_β):

$$F_\beta = \frac{(1 + \beta^2) \text{Precision} \times \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}},$$

$$\beta^2 = 0.3$$

3. **S-measure** (S_m), which evaluates structural similarity and preserves object-level integrity.

These metrics collectively capture pixel accuracy, region consistency, and structural fidelity.

D. Quantitative Results

Table 1 summarizes the performance of the proposed method on the ECSSD test set. Table 2 summarizes the performance of the proposed method on the DUTS-TE test set. The results demonstrate that our approach consistently outperforms baseline methods in both region-level and structure-level metrics. In particular, our model achieves a significantly lower MAE and higher F_β score, confirming its effectiveness in detecting fine-grained salient boundaries and complex object structures.

V. CONCLUSION

In this work, we presented an effective framework for salient object segmentation, leveraging a deep encoder-decoder architecture enhanced with region-aware and boundary-preserving mechanisms. The combination of Binary Cross-Entropy, MAE and Dice loss functions further improved the model’s ability to capture fine-grained details while maintaining global structural consistency. Trained on the DUTS-TR dataset and evaluated on the ECSSD benchmark, the proposed method demonstrated strong performance across multiple evaluation metrics, achieving accurate pixel-level predictions and robust generalization to diverse scenes. Qualitative and quantitative results consistently confirmed the effectiveness of our approach in handling challenging cases involving low contrast, cluttered backgrounds, and complex object shapes.

Some limitations in the framework. The model is susceptible to a loss of fine-grained spatial information. The reliance on MaxPool operations in the encoder, combined with patch-embedding convolutions in the transformer bottleneck, leads to an

aggressive reduction in spatial resolution. The learnable edge detection module, which operates directly on the raw input with a small kernel, is highly sensitive to noise. This module can lack the contextual scope required to effectively identify boundaries defined by complex texture changes.

Future work may involve integrating transformer-based attention modules, exploring multiscale feature aggregation strategies, and expanding the evaluation to larger and more diverse datasets. Additionally, extending the framework to video saliency or weakly supervised learning settings could offer promising directions for further research.

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Finally, the authors appreciate the contributions of previous researchers in salient object detection, whose foundational studies have laid out the groundwork for advancements in this field.

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